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Iowa State Fall 2020

EE 494 General Education Reflection

At Iowa State, undergraduate engineers are required to take general education credits in order to supplement their engineering knowledge with perspectives and information from other academic fields. I have not taken any general education courses at Iowa State as I graduated high school with enough AP course credits to also graduate from community college with an Associate degree. However, I took 2 honors seminars that, for the most part, fulfill the same intention. The honors seminars I took were Conflicts in the Middle East (equivalent to POL S 350) and Habits of the Mind (equivalent to PSYCH 310). For my other 2 general education courses, I will describe my experience in AP Government (POL S 215) and AP United States History (HIST 222).

My junior year of high school I took AP United States History which covered the events of the United States from the enlightenment period and colonization through present day. The course spent equal time teaching history as well as preparing us for the exam by practicing essay writing and analysis techniques. Learning United States history is important for every professional and every engineer because our society and educational backgrounds are a product of our previous generations. Learning about history reveals our country's problematic history and the contexts that allowed for those problems to arise. For example, learning about racial discrimination in the 20th century contextualizes present day diversity initiatives in STEM fields. Also, learning about the Great Depression and the Cold War contextualizes engineering booms during the New Deal and space race. History relates to engineering problem solving because problems need to be understood in their full context including their historical backgrounds.

My senior year of high school I took AP United States Government which covered current events, the organization of the state and federal governments, and a study of significant historical laws and judicial decisions. The course was taught during the 2016 Presidential Election, and I remember spending much of the class discussing current events rather than much a curriculum. Knowing about how the government runs is important both as a citizen and as an engineer. Public policy often shapes the way engineering standards and funding initiatives. Furthermore, engineering problem solving relates to public policy because working for a client in a heavily regulated environment is both how the government and engineering firms operate.

My freshman year of college I took an honors psychology seminar called Habits of the Mind. This course was about different mental habits and the way it shapes our behaviors. I remember discussions about habits such as persistence, thinking interpedently, and resisting temptations, among others. We would discuss they different ways one could implement these habits into their life and how each different type could change the way their brain operates. Understanding one's own mental behaviors and patterns is key to any high functioning engineer who needs to do their best to remain productive. Furthermore, understanding habits of the mind relates to engineering design thinking as it is crucial to be able to understand problems from a client's perspective and reorganize into problems for a team to solve.

My freshman year of college I took an honors political science seminar called Conflicts in the Middle East. This class was taught like a history class and each meeting we discussed a new major conflict, chronologically through history as each student would prepare a presentation for the week. Besides learning about important global history, this class had an emphasis on understanding bias in sources. Each conflict we studied was complex in its instigations and individual motivations, and often were radically different stories depending on what country was providing the information. We were tasked with discovering a neutral and complete understanding of each conflict, in addition to preparing an in-depth researched presentation. Learning about the inherent bias in sources is an important skill for any researcher since it demonstrates you understand evidence and the scenario better. Additionally, being a globally educated citizen relates to engineering problem solving as engineering problems often address the global population.

There are many problems facing society currently from environmental issues, global instability, political bipartisanism, economic inequality, and so forth. As engineers, we attempt to be problem solvers, and although the technology we develop might not directly impact on these large-scale problems, we would certainly like it to positively impact society in some way, and thus important for us to understand these issues. From my general education course, I learned about historical context of American problems, governmental operations, mental habits and behaviors for performance, and evaluating evidence for bias from a global perspective all which are important for solving future engineering problems.